GG-I-521
JUNE 28, 1960

FEDERAL SPECIFICATION

INDICATOR, STERILIZATION, (TABLET IN TUBE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of sterilization indicator, suitable for use in determining whether or not sterilizing temperature has been reached in the sterilizing chamber of autoclaves and sterilizers.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

CCC-T-191—Textile Test Methods.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

PPP-B-676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-L-10547 — Liners, Case, Waterproof.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Glass.*—Indicator tube shall be clear and colorless soda lime glass.

3.1.2 *Thread.* — Thread shall be black or white linen or cotton having a breaking strength of not less than 4 pounds. The dyed thread shall not bleed when tested as specified in 4.4.2.

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3.1.3 *Indicator tablet.* — Indicator tablet shall be as hereinafter specified. The tablet shall be solid material whose melting point shall be 248° ± 4°, −2°F., determined as specified in 4.4.1.

3.2 **Style and design.**—Sterilization indicator shall consist of a straight, heat-sealed, glass tube, 1¾ inches ± 1/8 inch in length and ¼ ± 1/32 inch in diameter. The tube shall contain the indicator tablet. The tablet shall contain a suitable dye; when the melting point is reached, a definite change in color shall occur, as an added indication of reaching the desired temperature. At one end of the tube, there shall be a constriction to which shall be secured a length of linen or cotton thread. The thread shall be of sufficient length so that when tied to the glass tube, there are two lengths of thread each not less than 12 inches long.

3.3 **Lot number.**—The manufacturer shall maintain records of control or lot numbers used on Government contracts (see 5.3).

3.4 **Count.**—The minimum average number of indicators per package shall be 100, with no less than 98 indicators in any individual package.

3.5 **Instructions.**—Full instructions for use shall accompany each unit package.

3.6 **Workmanship.**—Workmanship shall be first class throughout. Indicators shall be free from defects which detract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Supplier responsibilities for inspection.**—The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.*—When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 **Sampling.**—Unless otherwise specified, the unit of product for inspection purposes shall be the individual indicator.

4.2.1 *For examination.*—Sampling for examination shall be conducted in accordance with Military Standard MIL–STD–105.

	Inspection level	AQL (percent defective)
For visual examination:		
Major defects	II	1.0
Total (major and minor A combined) defects	II	4.0
Minor B (unit of product-box)	L1	6.5*
For dimensional examination	L1	2.5

*Not less than three samples shall be taken regardless of lot size.

4.2.2 *For tests.*—Sampling for tests shall be conducted in accordance with Military Standard MIL–STD–105.

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Component (unit of product)	Characteristic	Req.	Test procedure	Sample unit	Insp level	AQL
Indicator	Melting point—color change	3.1.3	4.4.1	Individual indicator	L1	*
Thread	Colorfastness	3.1.2	4.4.2	Cone or spool	L5	*
	Breaking strength	3.1.2	4.4.3			

*Acceptance number shall be zero, and not less than three samples shall be taken regardless of lot size.

4.3 **Examination.**—The indicators shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 *Classification of defects.*—Examination shall be conducted in accordance with the following classification of defects:

*Categories and defects**

Major

101. Glass tube not free of cracks and defective heat seals.
102. Glass tube not free of imperfections which will interfere with the observation of the indicator tablet.
103. Indicator tablet not included in tube.

Minor A

201. Indicator tube insufficiently constricted at one end to retain thread.
202. Thread insecurely fastened to tube.
203. Indicator tablet not in single pellet form.

*Minor B***

231. Count not within tolerances specified in 3.4.
232. Full instructions for use not included.

*Examination shall not be restricted to the classified possible defects listed above.

**Unit of product shall be unit of issue, and acceptance number shall be zero.

4.3.2 *Dimensional examination.*—The indicators shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4 **Tests.** — Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 *Determination of melting point.* — Melting point shall be determined by attaching the indicator tube to a suitable thermometer (calibrated to within ± 1°F. at 250°F.) in such a manner as to have the indicator tablet adjacent to the thermometer bulb. The thermometer, with indicator tube attached, shall be immersed in an oil bath (preheated to 212°F.), so that the indicator tube is completely immersed in the oil. The temperature of the oil shall be raised at the rate of not more than 2°F. per minute, and the oil shall be stirred to insure that the temperature throughout the bath is uniform at all times.

4.4.2 *Colorfastness of thread.*—A one-yard length of dyed thread shall be coiled and placed between layers of white gauze or cotton sheeting thoroughly moistened with distilled water and wrapped in dry huck towel and autoclaved for 20 minutes at 250°F., 15 pounds per square inch, and dried.

4.4.3 *Breaking strength.* — Breaking strength shall be determined in accordance with method 4100 of Federal Specification CCC–T–191.

5. PREPARATION FOR DELIVERY

(For civil agency procurement the levels of preservation, packaging, and packing used are defined in Federal Standard No. 102).

5.1 **Packaging.**

5.1.1 *Unit of issue.*—One hundred (100) indicators, as specified, constitute one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table I shall

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be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

TABLE I.—*Packaging quantities*

Unit package	Intermediate package	Exterior container
1 unit	12 units	144 units

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit shall be neatly bundled and secured with a suitable paper band or paperboard tube, and shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–676. Each carton shall be adequately secured to prevent accidental opening.

5.1.3.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP–B–636. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level B.*

5.1.4.1 *Unit package.*—One hundred (100) indicators as specified, constitute one unit. These one hundred indicators, with suitable cushioning material, shall be packaged in a set-up paperboard carton of appropriate size and design. Each carton shall be adequately secured to prevent accidental opening.

5.1.4.2 *Intermediate package.* — Twelve (12) unit cartons shall be packaged in an intermediate carton of appropriate size and design. Cartons shall be adequately sealed to prevent accidental opening.

5.1.5 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 **Packing.**

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table IV.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *For civil agencies.* — Twelve (12) intermediate cartons of twelve (12) units (144 boxes) shall be packed in a new shipping container. Carton shall be of appropriate size and design and shall be adequately secured to prevent accidental opening.

5.2.2.2 *For military agencies.* — Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table II or PPP–B–636, type I, class 1, fabricated from not less than 275-pound test fiberboard.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial con-

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tainers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *For civil agencies.*—Marking shall be in accordance with Federal Standard No. 123. Lot numbers shall be printed on unit packages.

5.3.2 *For military agencies.*

5.3.2.1 *Interior packages.*—Each unit package and each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129, except that date of manufacture shall be shown in lieu of date packed. Marking shall include lot (control) number.

5.3.2.2 *Exterior container.*—Exterior container shall be marked in accordance with Military Standard MIL–STD–129. Date of manufacture shall be shown in lieu of date packed. Lot (control) number shall be shown.

6. NOTES

6.1 **Ordering data.**—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.3 **Transportation description.**—Transportation description applicable to this item is:

Glassware, Chemical Laboratory, in boxes.
Carload minimum weight 18,000 pounds subject to Rule 34 of the Uniform Freight Classification.
Truckload minimum weight 18,000 pounds per National Motor Freight Classification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

CUSTODIANS:
Army—Army Medical Service
Navy—Bureau of Medicine and Surgery
Air Force.

Copies of this specification may be purchased for 5 cents each.

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